

A BAYESIAN SOLUTION TO THE BEHRENS-FISHER PROBLEM USING MONTE CARLO SIMULATION TECHNIQUES

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Key words: Bayesian solution; Behrens-Fisher problem; credibility region; exact moments; hypothesis testing; Monte Carlo simulation; multivariate case; Pearson curves.

Summary: Although it is difficult to derive the exact posterior distribution of a test statistic in the case of the Behrens-Fisher problem, the exact moments can be obtained in certain cases. By calculating the first four moments of the test statistic and by comparing Pearson curves to Monte Carlo simulation experiments, it is shown that good approximations of the true distribution can be obtained. Since there is no unique way to test hypotheses from a Bayesian point of view, Broemeling *et al.* (1990) proposed an approximate chi-square credibility region. In this note a "more exact" credibility region is derived. As mentioned, this is obtained by either using Pearson curves or Monte Carlo simulations. The theory and results are extended to the multivariate case.

1. Introduction

The literature devoted to the Behrens-Fisher problem is extensive and according to Broemeling *et al* (1990) there appears no satisfactory solution within

classical statistics. Because of the complexity of the "exact" solution, several approximate methods have been proposed and used (Smith, 1936; Welch, 1936, 1938 and Cochran & Cox, 1957). Wang (1971) showed that the two solutions by Welch, approximate degrees of freedom and a series solution now known as the Aspin-Welch solution (Aspin and Welch, 1949), are reasonably accurate in maintaining the derived probability of Type I error over a wide range of sample sizes and differences in variance.

Also Pfanzagl (1974) showed that, among the various competing methods, Welch's solution with approximate degrees of freedom tends to be uniformly most powerful against one-sided alternatives as the sample size increases. Therefore, it seems best to use Welch's solution involving approximate degrees of freedom. For further details about the Behrens-Fisher problem see also Barnard (1984) and Nel and van der Merwe (1986).

On the other hand, there is very little controversy within the Bayesian community and the Bayes solution for comparing two scalar normal means is reported *inter alia* by Box and Tiao (1973). Although the distribution is not exactly expressible in terms of tabled functions, a number of attempts have been made to approximate it in terms of tabled functions. Patil (1964) proposed a *t*-approximation by equating the second and the fourth moments. Recently Johnson and Weerahandi (1988) have proposed a Bayesian solution for the two-sample case. Their test statistic is a quadratic form that is similar to the one used by Broemeling *et al* (1990). They have determined the posterior distribution of the test statistic by one-dimensional numerical integration.

As mentioned by Broemeling *et al* (1990), the problem for the Bayesian is implementation. Since the exact posterior density is not a standard

distribution, the usual procedure is to use either numerical integration as was done by Box and Tiao (1973) and Johnson and Weerahandi (1988) or analytical approximation which is what Patil (1964) and Broemeling *et al* (1990) did.

In this note a third method namely that of Monte Carlo simulation will be introduced. Although it is difficult to derive the exact posterior distribution of a test statistic in the case of the Behrens-Fisher problem, the exact moments can be obtained in certain cases. By calculating the first four moments of the test statistic and by comparing Pearson curves to Monte Carlo simulation experiments, it will be shown that good approximations of the true posterior distribution can be obtained.

Since there is no unique way to test hypotheses from a Bayesian point of view, Broemeling proposed an approximate chi-square credibility region. In this note a "more exact" credibility region will be derived. As mentioned, this will be achieved by either using Pearson curves or Monte Carlo simulation.

In what follows both the terminology and layout will be along the same lines as those given by Broemeling *et al* (1990). Whereas they paid attention to approximations, our exposition will place more emphasis on the exact distributional aspects.

2. The Bayesian Formulation

Consider k independent normal populations $N(\mu_i, \sigma_i^2)$, where μ_i is the mean and σ_i^2 the variance of the i -th population, and $i = 1, 2, \dots, k$. The Behrens-Fisher problem is to test the hypothesis

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k. \quad (2.1)$$

Let $\{x_{ij} : j = 1, 2, \dots, n_i\}$ be a random sample of size n_i from the i -th population ($i = 1, \dots, k$). The first step in a Bayesian approach is to select some prior distribution. If one has little prior information, the Jeffreys prior density

$$p(\mu_1, \dots, \mu_k; \sigma_1^2, \dots, \sigma_k^2) \propto \prod_{i=1}^k \sigma_i^{-2} \quad (2.2)$$

is appropriate.

Combining the prior density (2.2) with the likelihood function, the joint posterior density function is given by

$$\begin{aligned} p(\mu_1, \dots, \mu_k, \sigma_1^2, \dots, \sigma_k^2 | X) &= p(\mu_1, \dots, \mu_k | \sigma_1^2, \dots, \sigma_k^2, X) p(\sigma_1^2, \dots, \sigma_k^2) \\ &= \left\{ \prod_{i=1}^k \left(\frac{2\pi\sigma_i^2}{n_i} \right)^{-\frac{1}{2}} \exp \left[-\frac{n_i}{2} (\bar{x}_i - \mu_i)^2 / \sigma_i^2 \right] \right\} \times \\ &\quad \left\{ \prod_{i=1}^k \left(\frac{(n_i - 1)}{2} s_i^2 \right)^{\frac{1}{2}(n_i - 1)} \frac{(\sigma_i^2)^{-\frac{1}{2}(n_i + 1)} \exp \left[-\frac{1}{2}(n_i - 1) s_i^2 / \sigma_i^2 \right]}{\Gamma \left(\frac{n_i - 1}{2} \right)} \right\} \end{aligned} \quad (2.3)$$

where

$$(n_i - 1) s_i^2 = \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2, \quad (2.4)$$

$$\bar{x}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} x_{ij} \quad (2.5)$$

and X is the "data matrix".

By eliminating the variances $\sigma_1^2, \dots, \sigma_k^2$ from (2.3) via integration, it can easily be shown (see for example Broemeling *et al*, 1990) that the k means are independent and that the posterior distribution of μ_i is a t -density with $n_i - 1$

degrees of freedom, mean $\bar{x}_i$ and dispersion

$$D_i = \frac{(n_i - 1) s_i^2}{n_i (n_i - 3)}, \quad n_i > 3. \quad (2.6)$$

Broemeling's solution to the Behrens-Fisher problem is based on approximating the t -density of μ_i with the normal approximation

$$\mu_i | X \sim N(\bar{x}_i, D_i) \quad (i = 1, \dots, k) \quad (2.7)$$

which has the same mean and variance as the actual posterior density.

As mentioned in the introductory paragraph our proposed solution is based on Monte Carlo simulations and approximations via Pearson curves.

In section 3 the posterior moments of a linear combination of μ_i 's will be derived and comparisons between exact and simulated posterior densities will be made. In section 4 the hypothesis (2.1): $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$ will be addressed while the multivariate case will be considered in section 5.

3. Monte Carlo Simulations and Pearson Curve Approximations

In theorem 3.1 the posterior moments of L (a linear combination of μ_i 's) are derived. Having obtained the moments, the Pearson curve can be obtained that best approximates the distribution of L .

Theorem 3.1

The mean, variance, third and fourth central posterior moments of $L = \sum_{i=1}^k a_i \mu_i$ are given by

$$\mu'_{1L} = \sum_{i=1}^k a_i \bar{x}_i \quad (3.1)$$

$$\mu_{2L} = \sum_{i=1}^k \frac{a_i^2 (n_i - 1) s_i^2}{n_i (n_i - 3)} \quad (3.2)$$

$$\mu_{3L} = 0 \quad (3.3)$$

$$\mu_{4L} = 3 \left\{ \sum_{i=1}^k \frac{a_i^4 (n_i - 1)^2 (s_i^2)^2}{n_i^2 (n_i - 3)(n_i - 5)} + 2 \sum_{i < j}^k \frac{a_i^2 a_j^2 (n_i - 1)(n_j - 1) s_i^2 s_j^2}{n_i n_j (n_i - 3)(n_j - 3)} \right\} \quad (3.4)$$

Proof. By first obtaining the unconditional posterior moments about zero and using the relationships between moments about the origin and central moments, the proof follows.

In section 3.1 illustrative examples are given and comparisons are made between exact, simulated and Pearson curve approximations of the posterior density.

3.1 Examples**Example 3.1**

The example below is called the "spinning modification experiment" and is given in Box and Tiao (1973, p. 107). (See also Broemeling *et al.*, 1990, p. 233.) The following sample data were obtained:

$$\bar{x}_1 = 50 \quad n_1 = 20 \quad s_1^2 = 12$$

$$\bar{x}_2 = 55 \quad n_2 = 12 \quad s_2^2 = 40$$

Table 3.1 compares the exact posterior density of Box and Tiao, the simulated and Pearson curve approximations as well as Broemeling's normal approximation (equation (2.7)) for $\delta = (\mu_2 - \mu_1)$. These densities are also displayed in figure 3.1.

Table 3.1. A comparison of the exact and approximate posterior densities of $\delta = \mu_2 - \mu_1$

δ	Exact density	Pearson density	Simulated density	Normal approx. (Broemeling)
5.0	0.1941	0.1930	0.1951	0.1832
5.2	0.1931	0.1919	0.1941	0.1824
5.4	0.1901	0.1889	0.1910	0.1801
5.6	0.1852	0.1840	0.1861	0.1762
5.8	0.1786	0.1773	0.1795	0.1710
6.0	0.1611	0.1691	0.1713	0.1647
6.2	0.1509	0.1597	0.1619	0.1572
6.4	0.1399	0.1493	0.1515	0.1489
6.6	0.1285	0.1383	0.1404	0.1397
6.8	0.1170	0.1269	0.1289	0.1301
7.0	0.1056	0.1154	0.1173	0.1202
7.2	0.0945	0.1041	0.1058	0.1100
7.4	0.0840	0.0931	0.0946	0.0997
7.6	0.0740	0.0826	0.0840	0.0898
7.8	0.0647	0.0728	0.0739	0.0801
8.0	0.0486	0.0637	0.0646	0.0709
8.4	0.0356	0.0478	0.0483	0.0541
8.8	0.0256	0.0351	0.0352	0.0400
9.2	0.0256	0.0253	0.0252	0.0285
9.6	0.0181	0.0179	0.0176	0.0197
10.0	0.0126	0.0125	0.0122	0.0131

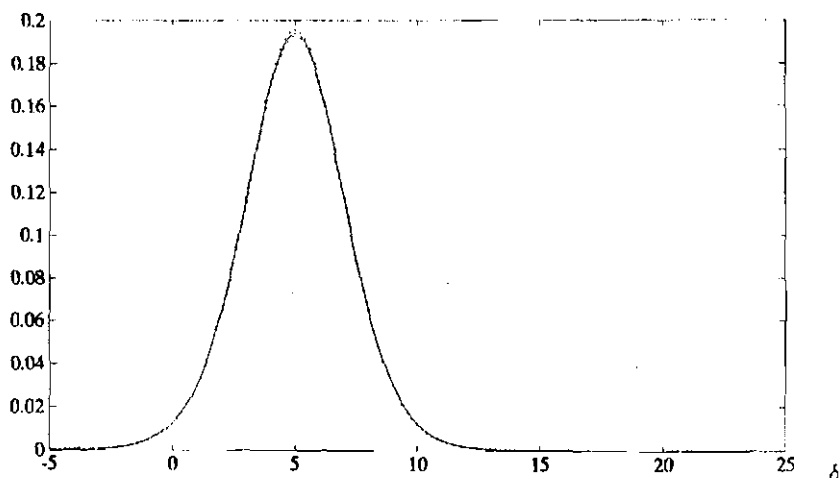


Figure 3.1. Exact and approximate densities of $f(\delta|X)$

(..... Simulation)

(- - - Exact)

(-.-.-. Pearson)

We see that the exact, simulated and Pearson curve densities are for all practical purposes the same but that Broemeling's normal approximation is "below" the exact density for values of $\delta = (\mu_2 - \mu_1)$ between 5.0 and 6.8 but after that stays above.

In table 3.2 the 95% HPD intervals for $\delta = (\mu_2 - \mu_1)$ are compared.

Table 3.2. Exact and approximate HPD intervals for $\delta = (\mu_2 - \mu_1)$

	Exact density	Pearson density	Simulated density	Normal approximation Broemeling
95% HPD Interval	0.673-9.327	0.685-9.339	0.660-9.314	0.731-9.269

From the table it can be seen that the Pearson and simulated intervals are nearly identical to the Box-Tiao interval but that the normal interval is shorter by 0.116.

The simulation study to approximate the posterior distribution of $\delta = (\mu_1 - \mu_2)$ was performed in the following way.

(i) From (2.3) it follows that

$$p(\sigma_i^2 \mid x_{i1}, \dots, x_{in_i}) = \left(\frac{(n_i - 1)s_i^2}{2} \right)^{-\frac{1}{2}(n_i - 1)} \\ \times \frac{(\sigma_i^2)^{-\frac{1}{2}(n_i + 1)} \exp \left[-\frac{1}{2}(n_i - 1)s_i^2 / \sigma_i^2 \right]}{\Gamma \left(\frac{n_i - 1}{2} \right)}$$

which is an Inverted Gamma distribution. The σ_i^2 ($i = 1, 2$) can therefore

be generated in the following way:

- (a) Draw z_i from a $\chi^2_{n_i-1}$ distribution.
- (b) Calculate $\sigma_i^2 = \frac{(n_i-1)s_i^2}{z_i}$ ($i = 1, 2$).
- (ii) Given σ_1^2 and σ_2^2 calculate the posterior density function $f(\delta | \sigma_1^2, \sigma_2^2, X)$, which is normal with mean $(\bar{x}_1 - \bar{x}_2)$ and variance $\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$.
- (iii) Repeat steps (i) and (ii) ℓ times. For these data ℓ was taken as 10 000. Using a Rao-Blackwell argument (see Gelfand and Smith, 1990), a density estimate of the unconditional posterior distribution of $\delta = (\mu_2 - \mu_1)$ can be obtained by averaging $f(\delta | \sigma_1^2, \sigma_2^2, X)$ over the ℓ repetitions.

For details of how to determine the parameters of a Pearson curve, given the values of its moments, see for example Elderton (1953) or Elderton and Johnson (1969). A type IV Pearson curve was used to approximate the posterior distribution of δ , the difference in means.

The density for this type of curve is given by

$$\tilde{f}(\delta) = K \left(1 + \frac{\delta^2}{a^2}\right)^{-m} e^{-v \tan^{-1}(\frac{\delta}{a})} \quad -\infty < \delta < \infty$$

where

$$r = \frac{6(\tilde{\beta}_2 - \tilde{\beta}_1 - 1)}{2\tilde{\beta}_2 - 3\tilde{\beta}_1 - 6};$$

$$m = \frac{1}{2}(r + 2);$$

$$v = \frac{-r(r-2)\sqrt{\tilde{\beta}_1}}{\{16(r-1) - \tilde{\beta}_1(r-2)^2\}} ;$$

$$a = \sqrt{\frac{\mu_2}{16}} \sqrt{16(r-1) - \tilde{\beta}_1(r-2)^2} ;$$

$$K^{-1} = \int_{-\infty}^{\infty} \tilde{f}(\delta) d\delta ;$$

$$K' = \frac{\tilde{\beta}_1(\tilde{\beta}_2 + 3)^2}{4(4\tilde{\beta}_2 - 3\tilde{\beta}_1)(2\tilde{\beta}_2 - 3\tilde{\beta}_1 - 6)} ;$$

with

$$\tilde{\beta}_1 = \frac{\mu_3^2}{\mu_2^3} \text{ and } \tilde{\beta}_2 = \frac{\mu_4}{\mu_2^2},$$

where $\mu'_1 = E(\delta)$; $\mu_2 = E\{\delta - E(\delta)\}^2$; $\mu_3 = E\{\delta - E(\delta)\}^3$ and $\mu_4 = E\{\delta - E(\delta)\}^4$.

These moments can be calculated from (3.1) – (3.4) by putting $a_1 = 1$, $a_2 = -1$ and $a_3 = a_4 = \dots = a_k = 0$.

By using numerical integration, the approximate percentage points of level $\alpha = 0.0025, 0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.75, 0.9, 0.95, 0.975, 0.99, 0.995$ and 0.9975 can be calculated.

Example 3.2

As another example of the proposed solution, consider a problem where we have three samples

$n_1 = 20$	$n_2 = 12$	$n_3 = 15$
$\bar{x}_1 = 50$	$\bar{x}_2 = 55$	$\bar{x}_3 = 60$
$s_1^2 = 12$	$s_2^2 = 40$	$s_3^2 = 50$

Suppose we want to test

$$H_0 : 2\mu_3 - \mu_2 - \mu_1 = 0 \quad \text{against}$$

$$H_a : 2\mu_3 - \mu_2 - \mu_1 \neq 0.$$

Define $\delta_1 = 2\mu_3 - \mu_2 - \mu_1$. Then, as in example 3.1, the unconditional posterior distribution of δ_1 can be obtained by simulation or Pearson curve approximation. In this case

$$\delta_1 | \sigma_1^2, \sigma_2^2, \sigma_3^2, X \sim N \left(2\bar{x}_3 - \bar{x}_2 - \bar{x}_1, \frac{4\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} + \frac{\sigma_3^2}{n_3} \right)$$

and $\sigma_i^2 (i = 1, 2, 3)$ can be generated as described in the previous example.

In figure 3.2 the simulated and Pearson curve (Type IV) approximations of $f(\delta_1|X)$ are illustrated.

From the figure it is clear that the two approximations of the posterior density of δ_1 are for all practical purposes the same. The 95% HPD interval for $\delta_1 = 2\mu_3 - \mu_2 - \mu_1$ computed from the approximate densities is given by (5.06 – 24.94). Since the 95% credibility interval does not include zero, H_0 is rejected.

4. The Solution to more Complicated Hypotheses in the Case of the Behrens-Fisher Problem

In order to test the hypothesis H_0 of the equality of means (2.1), Broemeling uses the marginal posterior distribution of $k - 1$ “independent” contrasts

$$\gamma_i = \mu_i - \mu_{i+1} \quad i = 1, 2, \dots, k - 1. \quad (4.1)$$

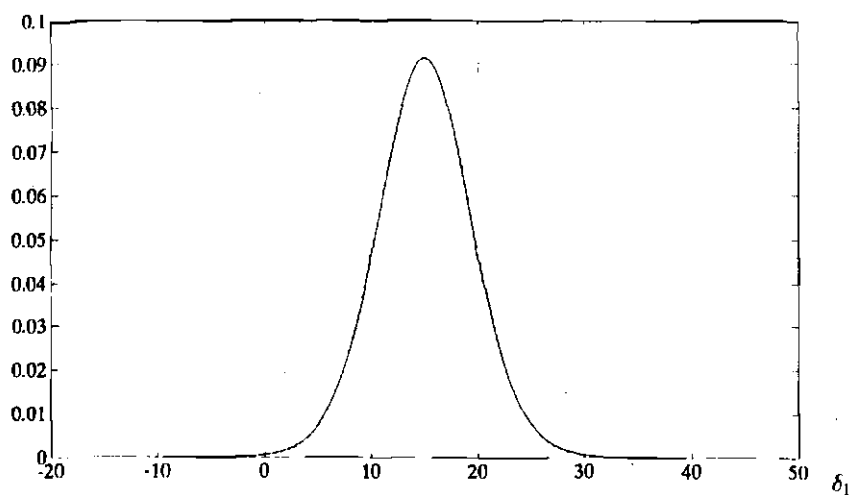


Figure 3.2. Pearson curve and simulated approximations of $f(\delta_1|X)$

(..... Simulation)

(---- Pearson)

Transforming from the k -means distribution (2.7) to the distribution of the $k - 1$ contrasts gives (according to Broemeling *et al*, 1990)

$$\gamma \sim N(\hat{\gamma}, D) \quad (4.2)$$

for the approximate posterior distribution of the contrast vector γ , where

$$\underline{\gamma} = [\gamma_1 \ \gamma_2 \ \dots \ \gamma_{k-1}]'; \quad \hat{\underline{\gamma}} = [(\bar{x}_1 - \bar{x}_2)(\bar{x}_2 - \bar{x}_3) \dots (\bar{x}_{k-1} - \bar{x}_k)]'$$

and the $(k - 1)$ order dispersion matrix

$$D = \begin{bmatrix} D_1 + D_2 & -D_2 & 0 & \dots & 0 \\ -D_2 & D_2 + D_3 & -D_3 & \dots & 0 \\ 0 & -D_3 & \dots & \dots & \vdots \\ \vdots & \dots & \dots & \dots & -D_{k-1} \\ 0 & \dots & \dots & -D_{k-1} & D_{k-1} + D_k \end{bmatrix} \quad (4.3)$$

where D_i is defined in (2.6).

A $(1 - \alpha)100\%$ approximate posterior credibility region for the vector $\underline{\gamma}$ of contrasts is given by

$$\left\{ \underline{\gamma} : (\underline{\gamma} - \hat{\underline{\gamma}})' D^{-1} (\underline{\gamma} - \hat{\underline{\gamma}}) \leq \chi_{\alpha, (k-1)}^2 \right\} \quad (4.4)$$

where $\chi_{\alpha, (k-1)}^2$ is the upper $100\alpha\%$ point of the chi-square distribution with $(k - 1)$ degrees of freedom; Broemeling *et al.* (1990) rejects the hypothesis

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

whenever

$$\hat{\underline{\gamma}}' D^{-1} \hat{\underline{\gamma}} > \chi_{\alpha, k-1}^2. \quad (4.5)$$

In this section a "more exact" credibility region will be derived. This will be achieved by using Monte Carlo simulation in the following way.

(i) Generate $\sigma_i^2 (i = 1, \dots, k)$ in the usual way.

(ii) $\underline{\gamma} | \sigma_1^2, \dots, \sigma_k^2, X \sim N(\hat{\underline{\gamma}}, \Sigma)$ where

$$\Sigma = \begin{bmatrix} \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} & -\frac{\sigma_2^2}{n_2} & 0 & 0 & \dots & 0 \\ -\frac{\sigma_2^2}{n_2} & \frac{\sigma_2^2}{n_2} + \frac{\sigma_3^2}{n_3} & -\frac{\sigma_3^2}{n_3} & 0 & \dots & 0 \\ 0 & -\frac{\sigma_3^2}{n_3} & \dots & \dots & \dots & \vdots \\ \vdots & \dots & \dots & \dots & \dots & -\frac{\sigma_{k-1}^2}{n_{k-1}} \\ 0 & \dots & \dots & -\frac{\sigma_{k-1}^2}{n_{k-1}} & \frac{\sigma_{k-1}^2}{n_{k-1}} + \frac{\sigma_k^2}{n_k} \end{bmatrix}$$

(iii) Given $\sigma_1^2, \dots, \sigma_k^2$ and the data, draw $\underline{\gamma}$ from $N(\hat{\underline{\gamma}}, \Sigma)$ and calculate

$$1. \Delta = (\underline{\gamma} - \hat{\underline{\gamma}})' D^{-1} (\underline{\gamma} - \hat{\underline{\gamma}}).$$

(iv) Repeat steps (i) – (iii) ℓ times and draw up the frequency table as well as the histogram of the calculated $\Delta_1, \dots, \Delta_\ell$ values.

Example 4.1

Consider again example 3.2 and suppose that we are interested in testing the hypothesis

$$H_0 : \mu_1 = \mu_2 = \mu_3 .$$

For $\ell = 10000$ the histogram of the simulated Δ values, the Pearson curve approximation of the posterior distribution of Δ (using the simulated moments) as well as Broemeling's approximation (χ^2_2) are shown in figure 4.1.

We see that the difference between the two densities (Pearson curve and χ^2_2) can be quite large – especially in the tails of the distribution.

In table 4.1 the percentage points of the two distributions are compared.

Table 4.1. Comparison of the percentage points for the two distributions of the test statistic $\Delta = (\underline{\gamma} - \hat{\underline{\gamma}})' D^{-1} (\underline{\gamma} - \hat{\underline{\gamma}})$

α	0.005	0.01	0.025	0.05	0.1	0.2
Pearson	0.008	0.017	0.043	0.082	0.185	0.398
χ^2_2	0.010	0.020	0.051	0.103	0.211	0.446
α	0.5	0.9	0.95	0.975	0.99	0.995
Pearson	1.288	5.089	6.319	8.103	9.457	10.899
χ^2_2	1.386	4.605	5.991	7.378	9.210	10.597

From the table it is clear that the percentage points of the Pearson distribution are smaller for values of α between 0.005 and 0.7 and larger for $0.75 \leq \alpha \leq 1$.

Also $\chi^2_{0.95,2} = 5.991$ while the 95% percentile of the Pearson distribution is 6.3189.

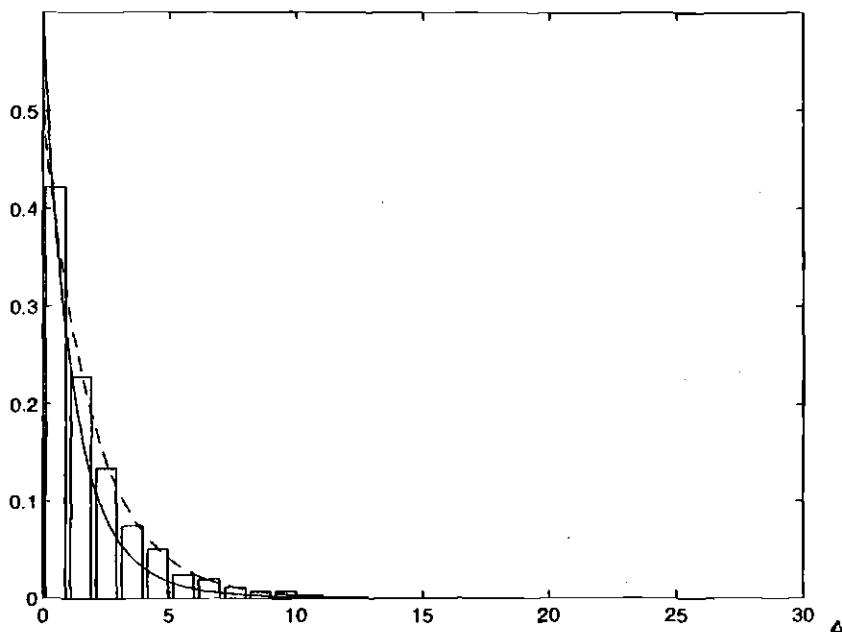


Figure 4.1.

Histogram of the simulated Δ -values

- Type VI Pearson curve approximation of the exact posterior distribution of the test statistic $\Delta = (\underline{\gamma} - \hat{\underline{\gamma}})' D^{-1} (\underline{\gamma} - \hat{\underline{\gamma}})$ using sample moments
- Broemeling's approximation (χ^2_2).

Weerahandi (1995) on the other hand considered a non-Bayesian solution to (2.1). He extended the representation of the two-sample test given in Tsui and Weerahandi (1989) to the one-way analysis of variance case and also provided exact tests for making multiple comparisons and for comparing variances. While his proposed test for $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$ is numerically equivalent to the test of Rice and Gaines (1989), it has a representation closely related to the conventional F-test and is computationally more efficient.

Weerahandi (1995) also established that the p -value appropriate for testing the null hypothesis (2.1) is indeed the exact probability of an unbiased and well defined subset of the sample space. This will not necessarily be the case for the Bayesian approach. The main advantage of a Bayesian approach, however, is that it allows explicit use of prior information thereby giving new insights in problems where classical statistics fail. Furthermore, under broader conditions, Bayesian estimators converge towards classical estimators when the sample size grows, if they are not identical.

5. The Multivariate Case

The simulation procedure described in the previous sections can easily be extended to the multivariate case. Therefore consider k independent p -dimensional normal populations $N(\underline{\mu}_i, \Sigma_i)$ where $\underline{\mu}_i$ is the mean vector and Σ_i the covariance matrix of the i -th population, where $i = 1, 2, \dots, k$. Suppose $\{\underline{x}_{ij} : j = 1, \dots, n_i\}$ is a random sample of size n_i from the i -th population. As in section 2, prior subjective uncertainty is described via the

"flat" prior

$$p(\underline{\mu}_1, \dots, \underline{\mu}_k; \Sigma_1, \dots, \Sigma_k) \propto |\Sigma_i|^{-\frac{1}{2}(p+1)} \quad (5.1)$$

Combining the prior density (5.1) with the likelihood function, the joint posterior density function is given by

$$p(\underline{\mu}_1, \dots, \underline{\mu}_k, \Sigma_1, \dots, \Sigma_k | X) = \left\{ \prod_{i=1}^k p_i(\underline{\mu}_i | \Sigma_i, X) \right\} \left\{ \prod_{i=1}^k p_i(\Sigma_i | X) \right\} \quad (5.2)$$

where

$$\underline{\mu}_i | \Sigma_i, X \sim N \left\{ \bar{\underline{x}}_i, \frac{1}{n_i} \Sigma_i \right\} \quad i = 1, \dots, k \quad (5.3)$$

and

$$p_i(\Sigma_i | X) \propto \frac{\exp \left\{ -\frac{1}{2} \text{tr} \tilde{S}_i \Sigma_i^{-1} \right\}}{|\Sigma_i|^{\frac{1}{2}(n_i+p)}}, \quad (5.4)$$

an Inverted Wishart Distribution.

Also

$$\tilde{S}_i = \sum_{j=1}^{n_i} (\underline{x}_{ij} - \bar{\underline{x}}_i)(\underline{x}_{ij} - \bar{\underline{x}}_i)' \quad (5.5)$$

and

$$\bar{\underline{x}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \underline{x}_{ij}$$

In what follows we will concentrate on the case where $k = 2$, that is two populations. The hypothesis of interest is therefore

$$H_0: \underline{\mu}_1 = \underline{\mu}_2$$

and the test statistic proposed by Broemeling *et al.* (1990) (see also Johnson and Weerahandi, 1988) is

$$Q = (\underline{\delta} - \underline{d})' D^{-1} (\underline{\delta} - \underline{d}) \quad (5.6)$$

where $\underline{\delta} = \underline{\mu}_1 - \underline{\mu}_2$, $d = \bar{x}_1 - \bar{x}_2$ and

$$D = \frac{\tilde{S}_1}{n_1(n_1 - 4)} + \frac{\tilde{S}_2}{n_2(n_2 - 4)}.$$

Broemeling approximated the posterior density of $\underline{\delta}$ by a bivariate normal distribution with mean $\underline{d}$ and dispersion matrix D and rejects H_0 at the $\alpha\%$ level whenever $\underline{d}' D^{-1} \underline{d} > \chi^2_{\alpha, p}$.

Johnson and Weerahandi (1988) proved that Q can be written as

$$Q = \underline{Z}' \left(\frac{\bar{V}_1}{n_1 L_1} + \frac{\bar{V}_2}{n_2 L_2} \right) \underline{Z}$$

where $\underline{Z}$ is distributed $N_p(\underline{0}, I)$, L_i is distributed $\chi^2_{n_i-p}$ for $i = 1, 2$, and the three variables are independent. Also $\bar{V}_i = D^{-\frac{1}{2}} \tilde{S}_i D^{-\frac{1}{2}}$. For exact computations see Johnson and Weerahandi (1988). In general, however, it is not easy to compute the exact distribution of Q analytically, but as mentioned its moments can be obtained or it can easily be simulated. The simulation procedure can be done in the following way:

- (i) Draw L_i ($i = 1, 2$) from a $\chi^2_{n_i-p}$ distribution and each component of $\underline{Z}$ from a univariate $N(0, 1)$ distribution.
- (ii) Calculate $Q = \underline{Z}' \left(\frac{\bar{V}_1}{n_1 L_1} + \frac{\bar{V}_2}{n_2 L_2} \right) \underline{Z}$.
- (iii) Repeat steps (i) and (ii) ℓ times and draw up a frequency table and histogram of the calculated $Q_1, \dots, Q_\ell$ values to display the unconditional posterior distribution of Q . $\ell = 10000$ will be appropriate.
- (iv) Calculate the 95% posterior credibility region for the vector $\underline{\delta}$. One way to do this is to order the 10000 simulated values of Q and use the 9500th rank value.

Example 5.1

Consider problem 7.6 of Press (1982, p. 201) (see also Broemeling *et al.*, 1990, p. 236) which gives

$$n_1 = 10 \qquad n_2 = 20$$

$$\bar{x}_1 = [4 \quad 7]' \qquad \bar{x}_2 = [6 \quad 6]'$$

$$\tilde{S}_1 = \begin{bmatrix} 10 & 3 \\ 3 & 8 \end{bmatrix} \quad \tilde{S}_2 = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}.$$

Using Broemeling's approximation, the posterior density of $\underline{\delta} = \underline{\mu}_1 - \underline{\mu}_2$ is bivariate normal with mean vector $\underline{d} = \bar{x}_1 - \bar{x}_2$ and dispersion matrix

$$D = \frac{\tilde{S}_1}{n_1(n_1 - 4)} + \frac{\tilde{S}_2}{n_2(n_2 - 4)} = \begin{bmatrix} 0.18541 & 0.05625 \\ 0.05625 & 0.14583 \end{bmatrix}.$$

The hypothesis of interest is that doubling the number of patrolling police officers does not have an effect on the average number of robberies (x_1) and on the average number of car thefts (x_2).

According to Broemeling this hypothesis will be rejected if

$$\underline{d}' D^{-1} \underline{d} > \chi_{0.05; 2}^2.$$

For this example the calculated value is 41.9014 and $\chi_{0.05; 2}^2 = 5.991$, thus the equality of means hypothesis ($H_0 : \underline{\mu}_1 = \underline{\mu}_2$) is rejected.

Using the simulation procedure we obtained the following:

For $\ell = 10000$ the histogram of the simulated Q values and Broemeling's approximation (χ_2^2) are shown in figure 5.1.

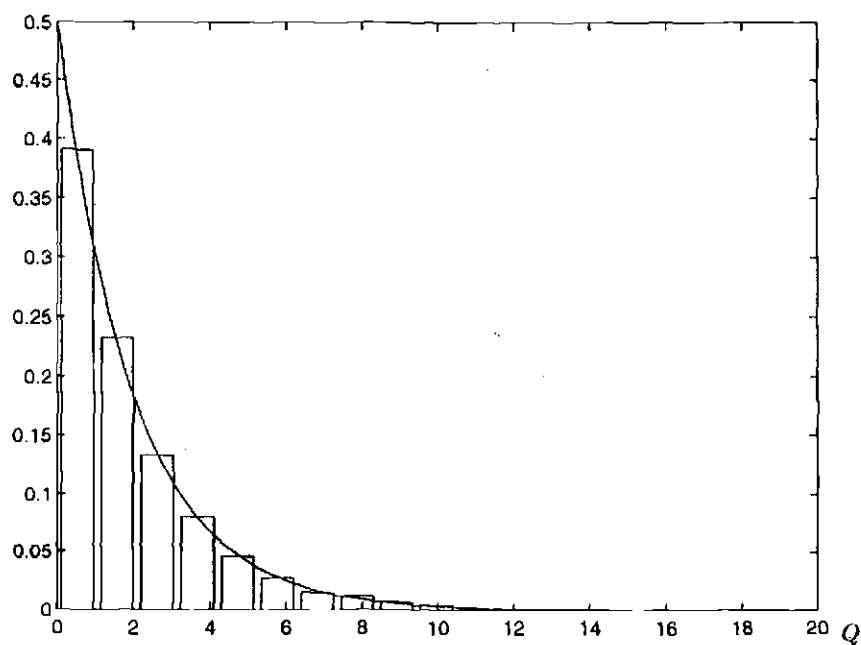


Figure 5.1. Histogram of the simulated Q -values
Broemeling's approximation (χ^2_2).

The 9500th rank value of Q was found to be 6.0213, which is larger than the $\chi^2_{0.05; 2} = 5.991$ and although 9.4248 (the 9900th rank value of Q) is larger than $\chi^2_{0.01; 2} = 9.21$ the equality of the means hypothesis is again rejected.

From this graph it again becomes clear that there is quite a difference between the two densities especially in the tails of the distribution.

For $k = 2$ it is also possible to obtain the exact moments of the test statistic Q defined in (5.6). This will be derived in theorem A.1 given in the appendix.

The testing procedure can also easily be extended to general k . Therefore consider the hypothesis

$$\begin{aligned} H_0 : \mu_1 = \mu_2 = \dots = \mu_k \\ \text{where} \end{aligned} \quad (5.8)$$

$$\mu_i (p \times 1) \quad (i = 1, 2, \dots, k).$$

In order to test this hypothesis the marginal posterior distribution of $k - 1$ "independent" contrasts

$$\underline{\delta} = \underline{\mu}_i - \underline{\mu}_{i+1} \quad i = 1, 2, \dots, k - 1$$

will be used.

Define

$$\underline{d}_i = \bar{x}_i - \bar{x}_{i+1},$$

$$\tilde{\underline{\delta}} = [\delta'_1 \quad \delta'_2 \quad \dots \quad \delta'_{k-1}]'$$

$$\tilde{\underline{d}} = [d'_1 \quad d'_2 \quad \dots \quad d'_{k-1}]'$$

$$\tilde{D}_i = \frac{\tilde{S}_i(n_i - p)}{n_i(n_i - p - 2)} \quad (p \times p)$$

and the $p(k - 1)$ order dispersion matrix

$$D = \begin{bmatrix} \tilde{D}_1 + \tilde{D}_2 & -\tilde{D}_2 & 0 & \dots & 0 \\ -\tilde{D}_2 & \tilde{D}_2 + \tilde{D}_3 & -\tilde{D}_3 & \dots & 0 \\ 0 & -\tilde{D}_3 & \dots & \dots & \vdots \\ \vdots & \dots & \dots & \dots & -\tilde{D}_{k-1} \\ 0 & \dots & \dots & -\tilde{D}_{k-1} & \tilde{D}_{k-1} + \tilde{D}_k \end{bmatrix}$$

Also $\tilde{\Sigma} p(k-1)$ is defined as

$$\tilde{\Sigma} = \begin{bmatrix} \frac{\Sigma_1}{n_1} + \frac{\Sigma_2}{n_2} & -\frac{\Sigma_2}{n_2} & 0 & 0 & \dots & 0 \\ -\frac{\Sigma_2}{n_2} & \frac{\Sigma_2}{n_2} + \frac{\Sigma_3}{n_3} & -\frac{\Sigma_3}{n_3} & 0 & \dots & 0 \\ 0 & -\frac{\Sigma_3}{n_3} & \dots & \dots & \dots & \vdots \\ \vdots & \dots & \dots & \dots & \dots & -\frac{\Sigma_{k-1}}{n_{k-1}} \\ 0 & \dots & \dots & -\frac{\Sigma_{k-1}}{n_{k-1}} & \frac{\Sigma_{k-1}}{n_{k-1}} + \frac{\Sigma_k}{n_k} \end{bmatrix}$$

From (5.3) it follows that

$$\tilde{\delta} | \tilde{\Sigma}, X \sim N(\underline{\tilde{d}}, \tilde{\Sigma}). \quad (5.9)$$

A test statistic for testing (5.8) can therefore be defined as

$$\tilde{Q} = (\tilde{\underline{\delta}} - \underline{\tilde{d}})' D^{-1}(\tilde{\underline{\delta}} - \underline{\tilde{d}}). \quad (5.10)$$

For general k , (5.10) cannot be written in a similar simplified form as given in (5.7) for $k = 2$. It is therefore very difficult, if not impossible, to derive the exact posterior distribution of $\tilde{Q}$ analytically. An estimate of the exact posterior distribution of $\tilde{Q}$ can, however, easily be obtained by using a simulation procedure. This can be done as follows:

- (i) From (5.4) it follows that $\Sigma_i^{-1} \sim W(\tilde{S}_i^{-1}, n_i - 1)$. It is therefore easy to simulate Σ_i^{-1} and therefore Σ_i . Simulation from the Wishart distribution can easily be done by using the algorithm of Odell and Feiveson (1966).
- (ii) Given $\Sigma_1, \dots, \Sigma_k$ and the data, draw $\tilde{\underline{\delta}}$ from a multivariate normal distribution with mean $\underline{\tilde{d}}$ and covariance matrix $\tilde{\Sigma}$. This follows from equation (5.9).
- (iii) Calculate $\tilde{Q} = (\tilde{\underline{\delta}} - \underline{\tilde{d}})' D^{-1}(\tilde{\underline{\delta}} - \underline{\tilde{d}})$.
- (iv) Repeat steps (i) – (iii) ℓ times and draw up a frequency table and histogram of the calculated $Q_1, \dots, Q_L$ values to display the unconditional posterior distribution of Q . $\ell = 10000$ will usually be appropriate.
- (v) Calculate the 95% posterior credibility region for the vector $\tilde{\underline{\delta}}$. As before, one way to do this is to order the 10 000 simulated values of $\tilde{Q}$ and use the 9500th rank value.
- (vi) Also reject the hypothesis (5.8) if $\underline{\tilde{d}}' D^{-1} \underline{\tilde{d}}$ is larger than the critical value

(9500th rank value).

6. Acknowledgements

The authors are grateful to the editor and the referee for their helpful suggestions and constructive comments which led to a substantial improvement of this paper.

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8. Appendix A – The moments of the test statistic Q .

Theorem A.1

Suppose $\underline{\mu}_i | \Sigma_i, X \sim N\left(\bar{x}_i, \frac{1}{n_i} \Sigma_i\right)$ where $\underline{\mu}_i (p \times 1)$ ($i = 1, 2$) and $\Sigma_i^{-1} \sim W(\tilde{S}_i^{-1}, n_i - 1)$. Define $\underline{\delta} = (\underline{\mu}_1 - \underline{\mu}_2)$, $\underline{d} = (\bar{x}_1 - \bar{x}_2)$ and $\tilde{V}_i = D^{-\frac{1}{2}} \tilde{S}_i D^{-\frac{1}{2}}$.

Also $D = \sum_{i=1}^2 \frac{\tilde{S}_i}{n_i(n_i - p - 2)}$ where $\tilde{S}_i$ is defined in (5.5).

The mean, variance, third and fourth central posterior moments of test statistic

$$Q = (\underline{\delta} - \underline{d})' D^{-1} (\underline{\delta} - \underline{d})$$

are given by

$$E(Q|X) = p \tag{A.1}$$

$$\begin{aligned}
 \text{Var}(Q | X) &= 2 \sum_{i=1}^2 \frac{1}{n_i^2(n_i - p - 2)(n_i - p - 4)} \\
 &\quad \left\{ \frac{(\text{tr} \bar{V}_i)^2}{(n_i - p - 2)} + \text{tr} \bar{V}_i^2 \right\} + \frac{4 \text{tr} \bar{V}_1 \bar{V}_2}{n_1 n_2 (n_1 - p - 2)(n_2 - p - 2)}
 \end{aligned}
 \tag{A.2}$$

$$\begin{aligned}
 \mu_3(Q | X) &= 8 \left\{ \sum_{i=1}^2 \frac{1}{n_i^3(n_i - p - 2)(n_i - p - 4)(n_i - p - 6)} \right. \\
 &\quad \left(\text{tr} \bar{V}_i^3 + \frac{3(\text{tr} \bar{V}_i)(\text{tr} \bar{V}_i^2)}{(n_i - p - 2)} + \frac{2(\text{tr} \bar{V}_i)^3}{(n_i - p - 2)^2} \right) + 3 \sum_{\substack{i=1 \\ i \neq j}}^2 \sum_{j=1}^2 \\
 &\quad \frac{1}{n_i^2 n_j (n_i - p - 2)(n_i - p - 4)(n_j - p - 2)} \\
 &\quad \left. \left(\text{tr} \bar{V}_i^2 \bar{V}_j + \frac{(\text{tr} \bar{V}_i)(\text{tr} \bar{V}_i \bar{V}_j)}{(n_i - p - 2)} \right) \right\}
 \end{aligned}
 \tag{A.3}$$

and

$$\begin{aligned}
 \mu_4(Q | X) &= \sum_{i=1}^2 \frac{12}{n_i^4(n_i - p - 2)(n_i - p - 4)(n_i - p - 6)(n_i - p - 8)} \\
 &\quad \left\{ 4 \text{tr} \bar{V}_i^4 + (\text{tr} \bar{V}_i^2)^2 + \frac{16(\text{tr} \bar{V}_i)(\text{tr} \bar{V}_i^3)}{(n_i - p - 2)} \right. \\
 &\quad \left. + \frac{2(n_i - p + 10)(\text{tr} \bar{V}_i)^2(\text{tr} \bar{V}_i^2)}{(n_i - p - 2)^2} \right\}
 \end{aligned}$$

$$\begin{aligned}
& + \frac{(n_i - p + 10)(tr \bar{V}_i)^4}{(n_i - p - 2)^3} \Big\} \\
& + \sum_{\substack{i=1 \\ i \neq j}}^2 \sum_{j=1}^2 \frac{12}{n_i n_j (n_i - p - 2)(n_i - p - 4)(n_j - p - 2)} \\
& \left\{ \frac{4}{n_i^2 (n_i - p - 6)} \left\{ 4tr \bar{V}_i^3 \bar{V}_j + (tr \bar{V}_i^2)(tr \bar{V}_i \bar{V}_j) + \frac{8(tr \bar{V}_i)(tr \bar{V}_i^2 \bar{V}_j)}{(n_i - p - 2)} \right. \right. \\
& + \left. \frac{(n_i - p + 2)(tr \bar{V}_i)^2 (tr \bar{V}_i \bar{V}_j)}{(n_i - p - 2)^2} \right\} + \frac{1}{n_i n_j (n_j - p - 4)} \\
& \left\{ 8(tr \bar{V}_i^2 \bar{V}_j^2) + 4tr(\bar{V}_i \bar{V}_j)^2 + 2(tr \bar{V}_i \bar{V}_j)^2 + (tr \bar{V}_i^2)(tr \bar{V}_j^2) \right. \\
& + \frac{16(tr \bar{V}_i)(tr \bar{V}_i \bar{V}_j) + 2(tr \bar{V}_i)^2 (tr \bar{V}_j^2)}{(n_i - p - 2)} \\
& \left. \left. + \frac{8(tr \bar{V}_i)(tr \bar{V}_j)(tr \bar{V}_i \bar{V}_j) + 12(tr \bar{V}_i)^2 (tr \bar{V}_j)^2}{(n_i - p - 2)(n_j - p - 2)} \right\} \right\}. \tag{A.4}
\end{aligned}$$

Proof. As mentioned, Johnson and Weerahandi (1988) proved that Q (equation (5.6)) can be written as

$$Q = \underline{Z}' \left(\frac{\bar{V}_1}{n_1 L_1} + \frac{\bar{V}_2}{n_2 L_2} \right) \underline{Z}$$

where $\underline{Z}$ is distributed $N_p(0, I)$, L_i is distributed $\chi_{n_i-p}^2$ for $i = 1, 2$, and the three variables are independent. Also $\bar{V}_i = D^{-\frac{1}{2}} \bar{S}_i D^{-\frac{1}{2}}$.

From Searle (1971, p. 55) it follows that the r -th cumulant of Q given L_1 and L_2 is

$$K_r(Q | X, L_1, L_2) = 2^{r-1}(r-1)!tr \left(\frac{\bar{V}_1}{n_1 L_1} + \frac{\bar{V}_2}{n_2 L_2} \right)^r \quad r = 1, 2 \quad (\text{A.5})$$

and from (A.5) the first four conditional moments about zero namely $\mu'_1(Q | X, L_1, L_2)$, $\mu'_2(Q | X, L_1, L_2)$, $\mu'_3(Q | X, L_1, L_2)$ and $\mu'_4(Q | X, L_1, L_2)$ can be obtained.

Since

$$E \left(\frac{1}{L_i} \right)^r = \frac{1}{(n_i - p - 2)(n_i - p - 4) \dots (n_i - p - 2r)}, \quad r = 1, 2, \dots$$

the unconditional posterior moments about zero can be derived after some lengthy algebra. Using the relationship between moments about the origin and central moments, equations (A.1) – (A.4) follow.

